# Jingfeng Wu

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## Academic Experience

**University of California, Berkeley** — Berkeley, US
Postdoctoral Fellow — 2023 - Present
Advisors: Peter Bartlett and Bin Yu

## Education

**Johns Hopkins University** — Baltimore, US
Ph.D. in *Computer Science* — 2019 - 2023
Advisor: Vladimir Braverman

**Peking University** — Beijing, CN
M.S. in *Applied Mathematics* — 2016 - 2019
B.S. in *Mathematics and Applied Mathematics* — 2012 - 2016

## Industrial Experience

**Google Research** — Seattle, US
Research Intern — Summer 2022
Mentors: Wennan Zhu and Peter Kairuoz

**Baidu Research** — Beijing, CN
Research Intern — Winter 2018
Mentor: Haoyi Xiong

## Research Interests

*Deep Learning Theory*, *Algorithms*, *Machine Learning*, *Optimization*, *Statistical Learning Theory*

## Preprints (dagger† indicates alphabetical order)

[1] **Jingfeng Wu**, Peter L. Bartlett †, Matus Telgarsky †, and Bin Yu †. "Large Stepsize Gradient Descent for Logistic Loss: Non-Monotonicity of the Loss Improves Optimization Efficiency". In: *arXiv preprint* (2024).

[2] Ruiqi Zhang, **Jingfeng Wu**, and Peter L. Bartlett. "In-Context Learning of a Linear Transformer Block: Benefits of the MLP Component and One-Step GD Initialization". In: *arXiv preprint* (2024).

## Conference Papers (asterisk* indicates equal contributions)

[3] **Jingfeng Wu**, Difan Zou, Zixiang Chen, Vladimir Braverman, Quanquan Gu, and Peter L. Bartlett. "How Many Pretraining Tasks Are Needed for In-Context Learning of Linear Regression?" In: *International Conference on Learning Representations (ICLR)* (2024).

[4] Xuheng Li, Yihe Deng, **Jingfeng Wu**, Dongruo Zhou, and Quanquan Gu. "Risk Bounds of Accelerated SGD for Overparameterized Linear Regression". In: *International Conference on Learning Representations (ICLR)* (2024).

[5] **Jingfeng Wu**, Vladimir Braverman, and Jason D. Lee. "Implicit Bias of Gradient Descent for Logistic Regression at the Edge of Stability". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[6] **Jingfeng Wu**, Wennan Zhu, Peter Kairouz, and Vladimir Braverman. "Private Federated Frequency Estimation: Adapting to the Hardness of the Instance". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2023).

[7] Haoran Li*, **Jingfeng Wu***, and Vladimir Braverman. "Fixed Design Analysis of Regularization-Based Continual Learning". In: *Conference on Lifelong Learning Agents (CoLLAs)* (2023).

[8] **Jingfeng Wu***, Difan Zou*, Zixiang Chen*, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "Finite-Sample Analysis of Learning High-Dimensional Single ReLU Neuron". In: *International Conference on Machine Learning (ICML)* (2023).

[9] **Jingfeng Wu***, Difan Zou*, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "The Power and Limitation of Pretraining-Finetuning for Linear Regression under Covariate Shift". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[10] Difan Zou*, **Jingfeng Wu***, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "Risk Bounds of Multi-Pass SGD for Least Squares in the Interpolation Regime". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2022).

[11] **Jingfeng Wu***, Difan Zou*, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "Last Iterate Risk Bounds of SGD with Decaying Stepsize for Overparameterized Linear Regression". In: *International Conference on Machine Learning (ICML)* (2022).

[12] **Jingfeng Wu**, Vladimir Braverman, and Lin F. Yang. "Gap-dependent Unsupervised Exploration for Reinforcement Learning". In: *International Conference on Artificial Intelligence and Statistics (AISTATS)* (2022).

[13] Difan Zou*, **Jingfeng Wu***, Vladimir Braverman, Quanquan Gu, Dean P. Foster, and Sham M. Kakade. "The Benefits of Implicit Regularization from SGD in Least Squares Problems". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[14] **Jingfeng Wu**, Vladimir Braverman, and Lin F. Yang. "Accommodating Picky Customers: Regret Bound and Exploration Complexity for Multi-Objective Reinforcement Learning". In: *Advances in Neural Information Processing Systems (NeurIPS)* (2021).

[15] Haoran Li, Aditya Krishnan, **Jingfeng Wu**, Soheil Kolouri, Praveen K. Pilly, and Vladimir Braverman. "Lifelong Learning with Sketched Structural Regularization". In: *Asian Conference on Machine Learning (ACML)* (2021).

[16] Zhuolong Yu, Chuheng Hu, **Jingfeng Wu**, Xiao Sun, Vladimir Braverman, Mosharaf Chowdhury, Zhenhua Liu, and Xin Jin. "Programmable Packet Scheduling with a Single Queue". In: *ACM Special Interest Group on Data Communication (SIGCOMM)* (2021).

[17] Difan Zou*, **Jingfeng Wu***, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "Benign Overfitting of Constant-Stepsize SGD for Linear Regression". In: *Conference on Learning Theory (COLT)* (2021).

[18] **Jingfeng Wu**, Difan Zou, Vladimir Braverman, and Quanquan Gu. "Direction Matters: On the Implicit Bias of Stochastic Gradient Descent with Moderate Learning Rate". In: *International Conference on Learning Representations (ICLR)* (2021).

[19] Jie You, **Jingfeng Wu**, Xin Jin, and Mosharaf Chowdhury. "Ship Compute or Ship Data? Why not Both?" In: *USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2021).

[20] Zhuolong Yu, **Jingfeng Wu**, Vladimir Braverman, Ion Stoica, and Xin Jin. "Twenty Years After: Hierarchical Core-Stateless Fair Queueing." In: *USENIX Symposium on Networked Systems Design and Implementation (NSDI)* (2021).

[21] **Jingfeng Wu**, Vladimir Braverman, and Lin F Yang. "Obtaining Adjustable Regularization for Free via Iterate Averaging". In: *International Conference on Machine Learning (ICML)* (2020).

[22] **Jingfeng Wu**, Wenqing Hu, Haoyi Xiong, Jun Huan, Vladimir Braverman, and Zhanxing Zhu. "On the Noisy Gradient Descent that Generalizes as SGD". In: *International Conference on Machine Learning (ICML)* (2020).

[23] Bing Yu*, **Jingfeng Wu***, Jinwen Ma, and Zhanxing Zhu. "Tangent-Normal Adversarial Regularization for Semi-Supervised Learning". In: *Conference on Computer Vision and Pattern Recognition (CVPR)* (2019).

[24] Zhanxing Zhu*, **Jingfeng Wu***, Bing Yu, Lei Wu, and Jinwen Ma. "The Anisotropic Noise in Stochastic Gradient Descent: Its Behavior of Escaping from Sharp Minima and Regularization Effects". In: *International Conference on Machine Learning (ICML)* (2019).

## Journal Papers (asterisk* indicates equal contributions)

[25] Difan Zou*, **Jingfeng Wu***, Vladimir Braverman, Quanquan Gu, and Sham M. Kakade. "Benign Overfitting of Constant-Stepsize SGD for Linear Regression". In: *Journal of Machine Learning Research (JMLR)* (2023).

## Professional Services

**Conference Reviewer**

- International Conference on Machine Learning (ICML) 2020 - 2024
- Conference on Neural Information Processing Systems (NeurIPS) 2020 - 2023
- International Conference on Learning Representations (ICLR) 2021 - 2024
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2021 - 2023
- Conference on Uncertainty in Artificial Intelligence (UAI) 2023

**Conference Program Committee Member**

- AAAI Conference on Artificial Intelligence (AAAI) 2021 - 2023

**Journal Reviewer**

- Journal of Machine Learning Research (JMLR)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- Transactions on Machine Learning Research (TMLR)
- Applied Probability Journals
- Journal of Artificial Intelligence Research (JAIR)

## Invited Talks

"Reimagining Gradient Descent: Large Stepsize, Oscillation, and Acceleration"

- **UC Berkeley**, *Biostatistics Seminar*, hosted by Lexin Li February 2024
- **UC San Diego**, *Group Seminar*, hosted by Mikhail Belkin February 2024

"New Insights about SGD: Stepsize, Risk Convergence, and Implicit Regularization"

- **UC Davis**, *Statistics Seminar*, hosted by Xiao Hui Tai Octorber 2023

"Implicit Bias of Gradient Descent for Logistic Regression at the Edge of Stability"

- **TTI-Chicago**, *MoDL Collaboration Meeting*, hosted by Sam Buchanan et al. May 2023

"The Implicit Regularization of SGD in Least Square Problems and Beyond"

- **Rice University**, *Algorithms and ML Seminar*, hosted by Anastasios Kyrillidis January 2023
- **Princeton University**, *Group Seminar*, hosted by Jason D. Lee December 2022
- **Georgia Tech**, *Group Seminar*, hosted by Molei Tao November 2022
- **Google Research**, *Learning Theory Seminar*, hosted by Mehryar Mohri August 2022
- **MPI & UCLA**, *Math Machine Learning Seminar*, hosted by Guido Montufar June 2022

## Teaching

- **Teaching Assistant** for "Machine Learning: Deep Learning", Johns Hopkins University Spring 2023

## Honors

- **Rising Stars in Data Science**, University of Chicago and UC San Diego 2023
- **MINDS Data Science Fellowship**, Johns Hopkins University Summer 2021
- **Best Reviewers (Top 10%)**, ICML 2021